UNIVERSIDAD DEL VALLE DE GUATEMALA

Facultad de Ingeniería
Redes
Semestre I – 2026



Laboratorio 03

Evidencias

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1. Topología

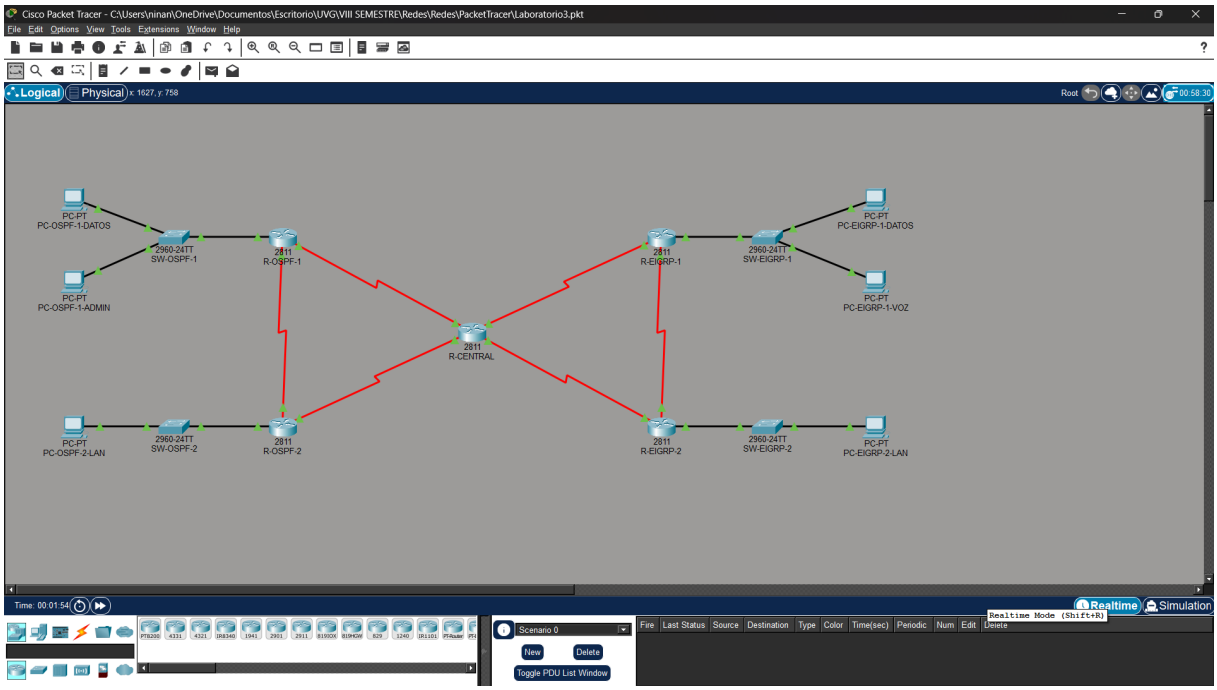


Gráfico 1: Topología de la red.

2. Tablas de enrutamiento

2.1. R-OSPF-1

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/1
L       10.0.0.1/32 is directly connected, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.5/32 is directly connected, Serial0/0/0
O       10.0.0.8/30 [110/20] via 10.0.0.6, 00:02:49, Serial0/0/0
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.6, 00:02:49, Serial0/0/0
O    192.168.0.0/24 [110/21] via 10.0.0.2, 00:02:39, Serial0/0/1
                    [110/21] via 10.0.0.6, 00:02:39, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, FastEthernet0/0.10
L       192.168.1.1/32 is directly connected, FastEthernet0/0.10
C       192.168.1.64/28 is directly connected, FastEthernet0/0.30
L       192.168.1.65/32 is directly connected, FastEthernet0/0.30

R-OSPF-1#
```

2.2. R-OSPF-2

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/1
L       10.0.0.2/32 is directly connected, Serial0/0/1
O       10.0.0.4/30 [110/20] via 10.0.0.10, 00:03:22, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/0
L       10.0.0.9/32 is directly connected, Serial0/0/0
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.10, 00:03:22, Serial0/0/0
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, FastEthernet0/0
L       192.168.0.1/32 is directly connected, FastEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/21] via 10.0.0.1, 00:03:22, Serial0/0/1
                       [110/21] via 10.0.0.10, 00:03:22, Serial0/0/0
O       192.168.1.64/28 [110/21] via 10.0.0.1, 00:03:22, Serial0/0/1
                        [110/21] via 10.0.0.10, 00:03:22, Serial0/0/0

R-OSPF-2#
```

2.3. R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/852224] via 10.0.0.18, 00:04:05, Serial0/0/0
D EX    10.0.0.4/30 [170/852224] via 10.0.0.18, 00:04:13, Serial0/0/0
D EX    10.0.0.8/30 [170/852224] via 10.0.0.18, 00:04:13, Serial0/0/0
C       10.0.0.12/30 is directly connected, Serial0/0/1
L       10.0.0.13/32 is directly connected, Serial0/0/1
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.17/32 is directly connected, Serial0/0/0
D       10.0.0.20/30 [90/849920] via 10.0.0.18, 00:04:11, Serial0/0/0
                     [90/1105664] via 10.0.0.14, 00:04:15, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.16.0.0/25 [90/284160] via 10.0.0.14, 00:04:15, Serial0/0/1
                      [90/852480] via 10.0.0.18, 00:04:11, Serial0/0/0
C       172.16.0.128/26 is directly connected, FastEthernet0/0.10
L       172.16.0.129/32 is directly connected, FastEthernet0/0.10
C       172.16.0.192/27 is directly connected, FastEthernet0/0.20
L       172.16.0.193/32 is directly connected, FastEthernet0/0.20
D    192.168.0.0/23 [90/1361664] via 10.0.0.18, 00:04:05, Serial0/0/0

R-EIGRP-1#
```

2.4. R-EIGRP-2

```
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/852224] via 10.0.0.22, 00:04:50, Serial0/0/0
D EX    10.0.0.4/30 [170/852224] via 10.0.0.22, 00:04:56, Serial0/0/0
D EX    10.0.0.8/30 [170/852224] via 10.0.0.22, 00:04:56, Serial0/0/0
C       10.0.0.12/30 is directly connected, Serial0/0/1
L       10.0.0.14/32 is directly connected, Serial0/0/1
D       10.0.0.16/30 [90/849920] via 10.0.0.22, 00:04:56, Serial0/0/0
                     [90/1105664] via 10.0.0.13, 00:05:00, Serial0/0/1
C       10.0.0.20/30 is directly connected, Serial0/0/0
L       10.0.0.21/32 is directly connected, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C       172.16.0.0/25 is directly connected, FastEthernet0/0
L       172.16.0.1/32 is directly connected, FastEthernet0/0
D       172.16.0.128/26 [90/284160] via 10.0.0.13, 00:05:00, Serial0/0/1
                        [90/852480] via 10.0.0.22, 00:04:56, Serial0/0/0
D       172.16.0.192/27 [90/284160] via 10.0.0.13, 00:05:00, Serial0/0/1
                        [90/852480] via 10.0.0.22, 00:04:56, Serial0/0/0
D    192.168.0.0/23 [90/1361664] via 10.0.0.22, 00:04:50, Serial0/0/0

R-EIGRP-2#
```

2.5. R-CENTRAL

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/30] via 10.0.0.9, 00:05:20, Serial0/0/1
                    [110/30] via 10.0.0.5, 00:05:20, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/281856] via 10.0.0.17, 00:05:28, Serial0/1/0
                     [90/281856] via 10.0.0.21, 00:05:26, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/28416] via 10.0.0.21, 00:05:26, Serial0/1/1
D       172.16.0.128/26 [90/28416] via 10.0.0.17, 00:05:28, Serial0/1/0
D       172.16.0.192/27 [90/28416] via 10.0.0.17, 00:05:28, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:05:20, Null0
O    192.168.0.0/24 [110/11] via 10.0.0.9, 00:05:20, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/11] via 10.0.0.5, 00:05:20, Serial0/0/0
O       192.168.1.64/28 [110/11] via 10.0.0.5, 00:05:20, Serial0/0/0

R-CENTRAL#
```

3. Balanceo OSPF

3.1. Interfaces OSPF de R-OSPF-1

```
R-OSPF-1#show ip ospf interface
FastEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
FastEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.65
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
```

```
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

3.2. Evidencias OSPF

```
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT TO POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

Gráfico 2: Costos de las interfaces seriales en R-OSPF-1.

```
R-OSPF-1#show ip route 192.168.0.10
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 21, type intra area
  Last update from 10.0.0.2 on Serial0/0/1, 00:13:13 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 00:13:13 ago, via Serial0/0/1
      Route metric is 21, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 00:13:13 ago, via Serial0/0/0
      Route metric is 21, traffic share count is 1

R-OSPF-1#
```

Gráfico 3: Dos rutas OSPF de costo igual hacia 192.168.0.0/24.

```
R-OSPF-1#show ip ospf interface

FastEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Gráfico 4: Interfaz LAN pasiva en OSPF.

4. Balanceo EIGRP

4.1. Tabla de topología de R-EIGRP-1

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.4/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.8/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.12/30, 1 successors, FD is 281600
         via Connected, Serial0/0/1
P 10.0.0.16/30, 1 successors, FD is 849664
         via Connected, Serial0/0/0
P 10.0.0.20/30, 2 successors, FD is 849920
         via 10.0.0.18 (849920/25856), Serial0/0/0
         via 10.0.0.14 (1105664/849664), Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 284160
         via 10.0.0.14 (284160/28160), Serial0/0/1
         via 10.0.0.18 (852480/28416), Serial0/0/0
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 1361664
         via 10.0.0.18 (1361664/537600), Serial0/0/0
R-EIGRP-1#
```

4.2. Evidencias EIGRP

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.4/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.8/30, 1 successors, FD is 852224
         via Rstatic (852224/28160)
P 10.0.0.12/30, 1 successors, FD is 281600
         via Connected, Serial0/0/1
P 10.0.0.16/30, 1 successors, FD is 849664
         via Connected, Serial0/0/0
P 10.0.0.20/30, 2 successors, FD is 849920
         via 10.0.0.18 (849920/25856), Serial0/0/0
         via 10.0.0.14 (1105664/849664), Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 284160
         via 10.0.0.14 (284160/28160), Serial0/0/1
         via 10.0.0.18 (852480/28416), Serial0/0/0
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 1361664
         via 10.0.0.18 (1361664/537600), Serial0/0/0
R-EIGRP-1#
```

Gráfico 5: Tabla de topología EIGRP y métricas de ambos caminos.

```
R-EIGRP-1#show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.13
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 4

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
 --More--
```

Gráfico 6: EIGRP AS 100 con variance 4.

```
R-EIGRP-1#show ip route 172.16.0.10
Routing entry for 172.16.0.0/25
Known via "eigrp 100", distance 90, metric 284160, type internal
  Redistributing via eigrp 100
  Last update from 10.0.0.18 on Serial0/0/0, 00:16:26 ago
  Routing Descriptor Blocks:
  * 10.0.0.14, from 10.0.0.14, 00:16:30 ago, via Serial0/0/1
      Route metric is 284160, traffic share count is 1
      Total delay is 10100 microseconds, minimum bandwidth is 100000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 1
    10.0.0.18, from 10.0.0.18, 00:16:26 ago, via Serial0/0/0
      Route metric is 852480, traffic share count is 1
      Total delay is 32300 microseconds, minimum bandwidth is 100000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 2

R-EIGRP-1#
```

Gráfico 7: Detalle de las métricas EIGRP hacia 172.16.0.0/25.

5. Redistribución y rutas de resumen

5.1. Configuración de R-CENTRAL

```
interface Serial0/1/0
 ip summary-address eigrp 100 192.168.0.0 255.255.254.0 5
!
interface Serial0/1/1
 ip summary-address eigrp 100 192.168.0.0 255.255.254.0 5
!
router eigrp 100
 redistribute ospf 1 metric 100000 10 255 1 1500
!
router ospf 1
 redistribute eigrp 100 metric-type 1
 redistribute static metric-type 1 subnets
!
ip route 172.16.0.0 255.255.255.0 Null0
```

5.2. Resumen de la configuración

La métrica semilla EIGRP es 100000 10 255 1 1500: ancho de banda (kbps), retardo (10 microsegundos), confiabilidad, carga y MTU. Se utiliza OSPF E1 para sumar el costo interno al externo.

Los resúmenes 192.168.0.0/23 y 172.16.0.0/24 agrupan las LAN y VLAN de OSPF y EIGRP, respectivamente.

Packet Tracer no admite summary-address en OSPF. Se redistribuye el agregado estático 172.16.0.0/24 a Null0 como E1; permanece aunque se pierdan las rutas EIGRP.

5.3. Evidencias de los resúmenes

```
R-EIGRP-1#show ip route 192.168.0.0
Routing entry for 192.168.0.0/23
Known via "eigrp 100", distance 90, metric 1361664, type internal
  Redistributing via eigrp 100
  Last update from 10.0.0.18 on Serial0/0/0, 00:17:39 ago
  Routing Descriptor Blocks:
  * 10.0.0.18, from 10.0.0.18, 00:17:39 ago, via Serial0/0/0
      Route metric is 1361664, traffic share count is 1
      Total delay is 52190 microseconds, minimum bandwidth is 100000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 1

R-EIGRP-1#
```

Gráfico 8: Resumen 192.168.0.0/23 recibido en el dominio EIGRP.

```
R-OSPF-1#show ip route 172.16.0.10
Routing entry for 172.16.0.0/24
Known via "ospf 1", distance 110, metric 30, type intra area
  Last update from 10.0.0.6 on Serial0/0/0, 00:15:22 ago
  Routing Descriptor Blocks:
  * 10.0.0.6, from 5.5.5.5, 00:15:22 ago, via Serial0/0/0
      Route metric is 30, traffic share count is 1

R-OSPF-1#
```

Gráfico 9: Resumen 172.16.0.0/24 recibido en el dominio OSPF.

6. Prueba de conectividad extremo a extremo

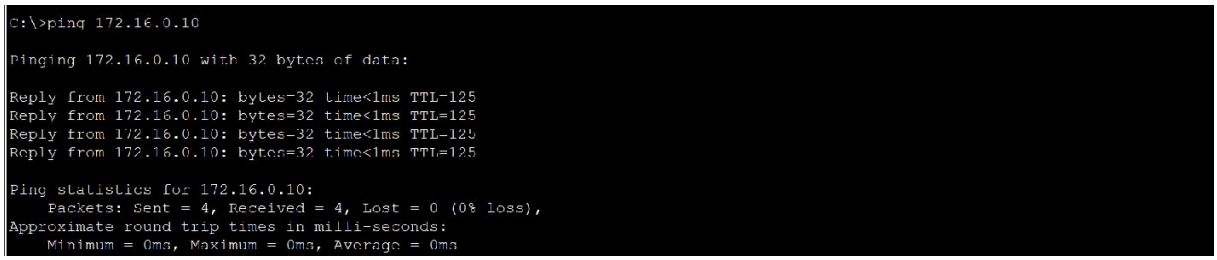


Gráfico 10: Ping: 192.168.1.10 a 172.16.0.10.

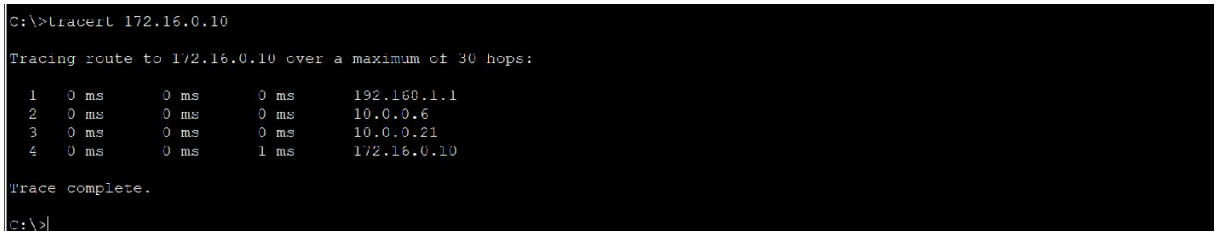


Gráfico 11: Tracert: 192.168.1.10 a 172.16.0.10.

7. Parámetros del balanceo

OSPF: ambos caminos tienen costo 20 + 1 = 10 + 10 + 1 = 21.

EIGRP: las métricas 284160 y 852480 tienen relación 1:3, correspondiente al objetivo teórico 75/25. La alternativa cumple factibilidad (28416 < 284160) y se instala con variance 4.